

Interfacing Children and Robots for Stress Management in ASD Therapy

María Gaitán-Padilla¹ *IEEE Member*, Maria José Pontes¹, Carlos A. Cifuentes² *IEEE Senior Member* and Camilo A. R. Diaz¹ *IEEE Member*

Abstract—Children with autism spectrum disorder (CwASD) are highly exposed to social stressors yet often struggle to recognize or communicate their own stress. Timely and accurate physiological feedback may help prevent erratic behaviours related to emotional regulation, such as self-injury, while enhancing therapeutic engagement. To address this, an affordable ecosystem of social assistive technologies is being developed that combines a low-cost social robot with wearable optical fibre sensing. The robot, designed for accessibility and adaptability, has been implemented in therapy sessions with children with ASD, showing positive effects on attention and sustained engagement. In parallel, an optical fibre sensor for heart rate variability (HRV) detection was developed and validated: first with adults, then with a case study involving a child with ASD. Building on these results, a biofeedback protocol was applied using a personalized algorithm during a speech therapy session of alternative communication training. Preliminary results from a single case indicate reduced time in stress states and fewer disruptive behaviours during the robot-mediated session compared to the human-assisted session. Future studies will expand the number of sessions and participants to further evaluate the influence of robot-mediated biofeedback in alternative communication therapies.

Index Terms—Autism Spectrum Disorder, Biofeedback, Child-Robot Interaction, Stress Detection.

I. BACKGROUND AND RELATED WORK

Stress is commonly assessed through bodily responses associated with autonomic nervous system (ANS) activation. Variables such as heart rate (HR) [1], respiratory rate (RR) [2], blood pressure (BP) [3], cortisol levels in saliva [4], and galvanic skin response (GSR) [5] have been widely used as physiological indicators. In parallel, psychosocial assessments capture stress-related parameters such as pupil dilation, negative affect, appetite changes, and sleep disturbances, with self-evaluation remaining central to most anxiety and stress approaches [6]. Among these variables, HR and RR have shown strong correlations with stress [1]. GSR, speech, and sweating are also informative, with electrodermal activity identified as one of the most consistent ANS responses to stressors [5]. Importantly, studies in populations with neurodevelopmental disorders, including

ASD, have explored stress induction protocols to characterize differences in stress responses relative to typically developing (TD) children [6].

For physiological monitoring, optical fibre sensors have gained prominence, particularly Fibre Bragg Grating (FBG) sensors due to their multiplexing capabilities [7]. These have been used to measure cardiac pulse signals at multiple body sites, including the radial (wrist) and brachial (elbow) arteries, as well as the chest. For instance, Koyama *et al.* [1] validated an FBG-based HR sensor against an ECG device (Nihon Kohden) and a sphygmomanometer, while Tavares *et al.* [8] demonstrated simultaneous HR and RR monitoring with a single FBG system. However, despite their accuracy, FBG sensors remain limited by their bulk and costly interrogation systems. To address this, polymeric optical fibre (POF) sensors have emerged as a cost-effective alternative. These sensors exploit optical power variations and have been used to measure temperature, strain, HR, HRV, and RR in dynamic scenarios [9]. Sensor positioning has also been studied to optimize physiological signal acquisition, and their flexibility and small size have enabled integration into wearable and textile-based systems, with applications in gait analysis and activity recognition [10].

Anxiety management strategies often include breathing and relaxation exercises prescribed by therapists. Yet, children with ASD (CwASD) frequently struggle to identify moments of heightened anxiety or stress. To address this, Nguyen *et al.* [2] developed a feedback system using a commercial wearable (Shimmer, Shimmer Sensing Ltd.), prompting users to perform breathing and relaxation exercises during stress episodes. The intervention group outperformed a control group that received only tablet-based instructions. Similarly, Melo *et al.* [11] studied Robot-Assisted Therapy (RAT) for CwASD using wearable physiological sensors to capture ANS responses during Child-Robot Interactions (CRIs). In other Human-Robot Interaction (HRI) functionalities, sensing technologies are typically used to characterize or control robot behaviour, including pressure and touch sensors [12], as well as vision-based systems for environmental perception [13]. However, optical fibre sensors have not yet been applied in social robotics. Building on these works, the present project advances research on social robotics for ASD therapy [14], [15] by developing low-cost optical fibre sensors for stress detection in CwASD [16] and instrumenting a social robot to provide feedback during speech therapy sessions [17].

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¹ Telecommunications Laboratory (LabTel), Electrical Engineering Department, Federal University of Espírito Santo, Vitória, Brazil maria.padilla@edu.ufes.br (maria.pontes, camilo.diaz)@ufes.br

² Bristol Robotics Laboratory, University of the West of England, Bristol, UK. carlos.cifuentes@uwe.ac.uk

II. RESEARCH QUESTION

Based on the introduction, this project works on the following research questions:

- Can stress in children with ASD be reliably detected using the developed POF-based wearable?
- Does providing children with physiological feedback help prevent stress-related erratic behaviours?
- Can a social robot effectively deliver stress feedback and adaptive support during therapy sessions?

To address these questions, we developed a low-cost optical fibre sensor system to measure stress through ANS-related physiological responses in CwASD. Building on this, we proposed a personalized biofeedback system that leverages the motivational role of RAT in speech therapy sessions. In addition, the robot is being instrumented with optical fibre touch sensors to classify child-robot physical interactions, further enriching engagement analysis and adaptive support.

III. METHODOLOGY

The research was structured into four stages to address the proposed research questions, building on ongoing work in CRI during ASD therapies, presented in Figure 1. Starting with the development of the optical fibre sensor, as a low-cost wearable sensor made of POF for online measurement of physiological parameters associated with stress detection (breathing rate, heart rate, and heart rate variability). Then, the stress detection validation was made, recording the HRV metrics under rest and stress conditions in a TSST. Data collection was complemented by comparisons with standard commercial sensors and databases for stress estimation [18]. After that, therapists and educators from the partner rehabilitation centre for children with ASD collaborated in designing the therapeutic protocols, ensuring safety, clinical appropriateness, and ecological validity. Special attention was given to managing CRI and tailoring stress-related stimuli and feedback to individual needs. In this sense, the social robot was introduced in institutional speech therapies for social skills and in-group sessions, controlled by the Wizard of Oz (WoZ) method. The goal was to integrate the robot in the traditional activities of the school, adapted to their needs. Finally, the co-created personalized biofeedback protocol was conducted with six children with ASD. This work presents one case study as a preliminary result, integrating wearable sensor data with robot-mediated feedback compared with a teacher giving the biofeedback during a speech therapy session focused on alternative communication. The project thus integrates wearable stress sensing, social robotics, and participatory protocol design to create a clinically grounded system for adaptive stress management in ASD therapy.

A. Robotic Platform

The CASTOR robot¹ is an affordable, open-source platform designed for therapies of children with ASD. To ensure the safety of users, it has elastic actuators and bioinspired tendons. The robot is equipped with seven motors

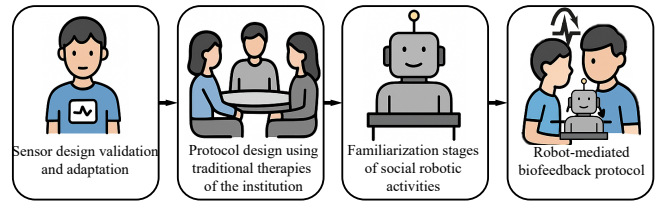


Fig. 1: Project's pipeline of sensor development, protocol co-creation, institution familiarization to the social robot, and the proposed protocol for biofeedback.

(Dynamixel MX/AX, Robotis, France), five servomotors (MG996, Towerpro, Taiwan), and a passive damping system resulting in 14 degrees of freedom. Besides, CASTOR is a robust 3D-printed modular device, easy to modify, and repair. This robot was implemented in ASD therapies working on the performance of social skills affected in CwASD and became a motivational tool in the conducted therapies by obtaining more than 1000 spontaneous physical interactions from the participating children in the study [15]. In addition, the robot has been validated as a huggable [19] and robust robot against different physical interactions [14], which can be helpful in emotional control therapies in CwASD.

B. Physiological sensor

The POF sensor was built using SH4001 Simplex fiber (PMMA core, NA = 0.5, 980/1000 μm core/cladding, 2200 μm polyethylene jacket). The system operates on optical intensity variation, a constant LED source couples light into the fiber, and a photodetector measures changes in intensity caused by chest expansion and contraction from pulse and respiration. A custom 3D-printed piece holds one fiber end fixed to the LED and the other to the photodetector, with the gap modulated by thoracic motion [18]. Two sensing points were embedded in an elastic Velcro band to provide redundancy. The photodetectors were connected to a 24-bit AD/DA board (Waveshare) and a single-board computer (Raspberry Pi), powered by a portable battery. Data were stored locally and transmitted via Wi-Fi (MQTT) to a computer running Matlab at 120 Hz. The electronics were enclosed in a 3D-printed case carried in a small backpack.

C. Experimental Procedure

The participant was selected based on professional recommendations, selecting a six-year-old child being part of the training of the picture exchange communication system (PECS), with sensory processing profiles and language delay. The PECS session was conducted twice, with the robot and with an assistant teacher conducting the biofeedback, according to the physiological states (Figure 2). The session starts with a brief introduction to the session and the presence and role of the robot/teacher. During the introduction, the session was explained in its three phases to the participant, presenting the elements (the bird-box game, crayons, the PECS board, and the clock). The session started with a 3-minutes bird-box game, where the participant had to match

¹<https://github.com/CastorProject/CASTOR-Robot/wiki>

the color of the birds with the boxes. Then the PECS activity was conducted, where crayons were used as reinforcers to motivate communication intention using the pictogram exchange with the speech therapist, who controlled the access to the reinforcers. Finally, free-play time with the crayons was provided, generating calm after the PECS request task. During both sessions, the biofeedback was conducted using the sensor classification and predetermined cues for the three physiological states (i.e., for stress stop and promote breathing, for regular activity motivation phrases to continue, and for relaxation, congratulate the participant).

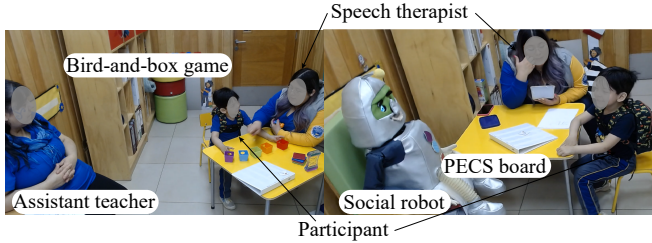


Fig. 2: PECS session setup. a) Teacher as the social agent for biofeedback. b) Robot as the social agent providing biofeedback.

To complement the physiological classification and understand participant behaviour during the sessions, video analysis was conducted for both sessions. Two sets of variables were extracted, the distribution of physiological states over time and the therapeutic behaviours and events: (1) the number of biofeedback interventions, (2) the number of correct PECS communications, (3) the number of incorrect PECS communications, and (4) the number of disruptive behaviours. Together, these video-based variables provide a behavioural perspective to complement the physiological classification, enabling a multidisciplinary evaluation of the biofeedback framework during PECS-based therapy.

IV. PRELIMINARY RESULTS

Figure 3 illustrates the distribution of time spent in three physiological states (Relax, Activity, Stress) during PECS sessions with the CASTOR robot and with the assistant teacher. The comparison suggests that in the robot-mediated session the child spent more time in Relax and Activity states, while the Teacher-supported session showed a higher proportion of time in Stress. This trend indicates a potential reduction in stress when biofeedback was delivered by the robot.

Therapeutic behaviour metrics further support these observations. In the robot-assisted session, nine feedback events were delivered and no disruptive behaviours occurred. In the teacher-supported session, eight feedback events were provided, but two disruptive behaviours were recorded. Regarding PECS performance, the child completed nine correct requests with one error in the robot-assisted session, compared to eight correct requests and three errors in the teacher session.

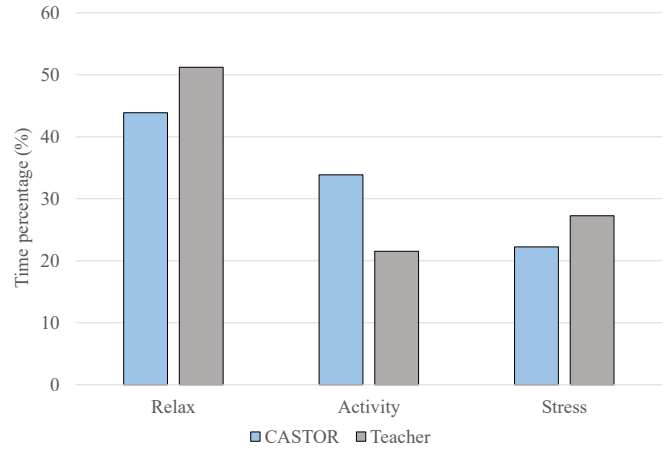


Fig. 3: Preliminary results of physiological states in the robot-mediated session and the teacher-guided feedback.

Together, these findings provide preliminary evidence that robot-mediated feedback may not only influence the child's physiological state distribution but also reduce disruptive behaviours and improve accuracy in PECS performance.

V. CONCLUSION AND FUTURE WORK

These preliminary findings suggest that integrating physiological sensing with robot-mediated feedback can provide valuable support during PECS therapies for CwASD. Analysis of physiological state distributions shows a reduction in time spent under stress and an increase in regular activity during the robot-assisted session in this case study. In addition, session metrics point toward better engagement and performance, with fewer disruptive behaviours and slightly higher accuracy in PECS requests when the CASTOR robot was present compared to sessions supported by the assistant teacher. Although based on this preliminary case, the results highlight the possibility of testing the CASTOR robot within more robot-mediated feedback, which may positively influence both stress regulation and therapeutic performance. Further studies with larger cohorts and long-term validation are required to confirm these trends, refine the biofeedback mechanisms, and better understand the role of CRI in supporting emotional regulation and learning outcomes in ASD therapies.

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